

Global Organ Transplantation

Regional disparities, donor sources and the COVID stress test

Alessandro Capoccia | R portfolio project | Data period 2000–2024

Overview

The starting point of this project was a question about differences in organ donation across countries. The Global Observatory on Donation and Transplantation (GODT) dataset provides a way to examine these differences through donation rates, donor composition and utilization. The analysis also considers the disruption in 2020 and the extent of recovery in 2021.

The main finding is that comparisons depend on both donation activity and reporting coverage. Regional differences are substantial, and most countries in the COVID sample remained below their 2019 rate in 2021. Two apparent breaks in the historical series also show why a change in an aggregate should be checked against the countries contributing to it.

Data cleaning and tidying

The source workbook contains 4,645 country-year records for 194 countries over 2000–2024, with 20 donation and transplantation indicators. We reshaped these indicator columns into a long format, with one row for each country, year and indicator. The full table has 92,900 rows; 32,373 contain reported values. A country listed in the workbook does not necessarily report each indicator.

For the portfolio version, country names were standardised and the population conversion was corrected. Commas used as thousands separators in China's and India's population entries had prevented numeric conversion. Population is recorded in millions, so dividing a donor count by this field gives donors per million population (pmp). Non-positive population values were excluded from rate calculations.

Missing values were kept separate from reported zeros. The checks found no duplicate country-year-indicator keys. Differences between reported totals and their components were flagged and retained for review. Regional donor rates use the median across reporting countries; donor shares and utilization ratios use the denominators described in each section. These are descriptive comparisons, without a causal model.

Project contribution

The original group consisted of Alessandro Capoccia, Alessandro Faglia, Federico Garagnani and Enrico Telio. My contribution focused on regional disparities, the COVID stress test and living-versus-deceased donation. Faglia led data tidying and the utilization analysis; Garagnani analysed DBD and DCD; Telio wrote the report with Garagnani's support. The work also involved collaboration across sections.

I later developed the R section labelled "Additional Personal Analysis". This independent extension examines recovery in 2021, regional DCD shares, country-level utilization and reporting coverage. The following sections distinguish these additions from the original group analysis.

Regional disparities and reporting coverage

We first compared the median actual deceased donor rate across regions. Europe has the highest median from 2004 onwards, reaching 15.7 pmp in 2024. The other regional medians remain below 5 pmp in that year. Each country receives equal weight, so these values describe the typical reporting country rather than a population-weighted regional rate.

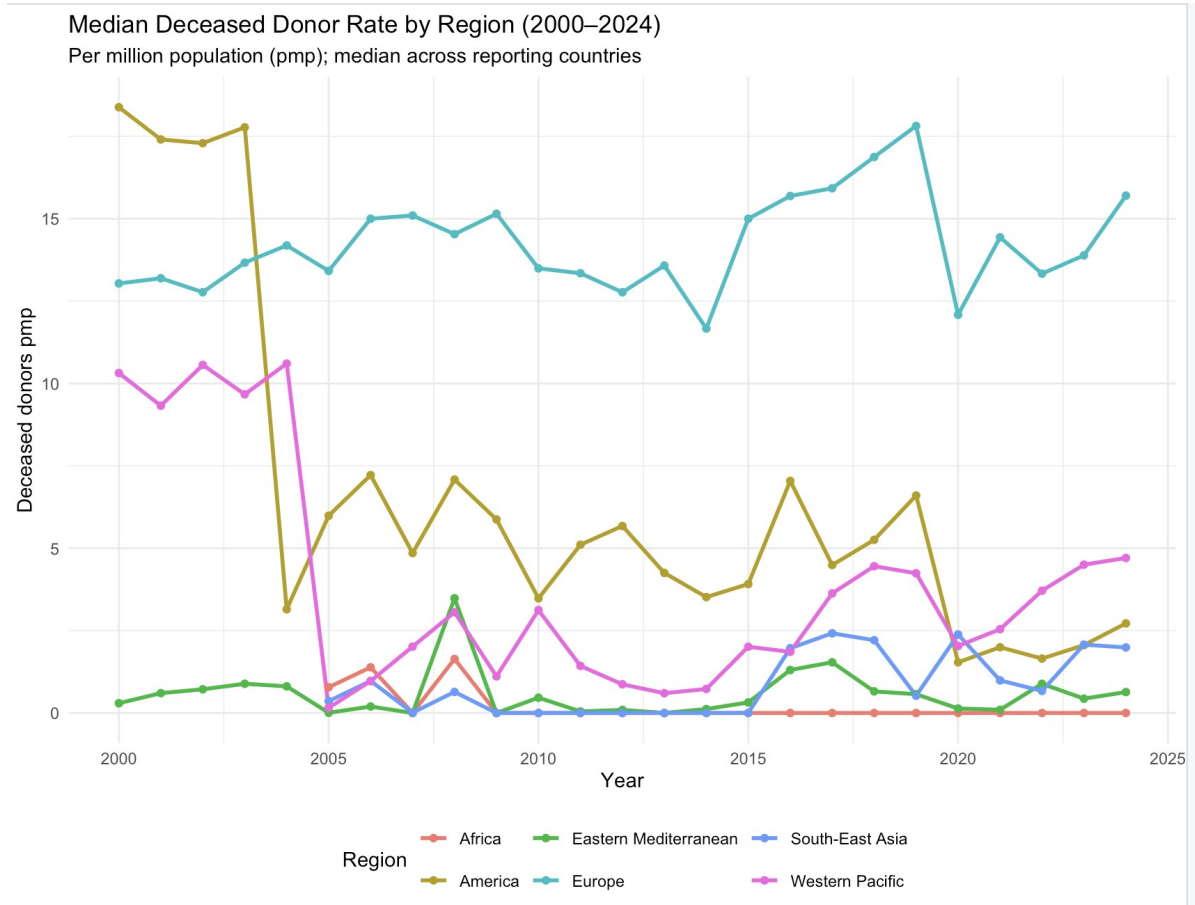


Figure 1. Median actual deceased donor rate by region, 2000–2024. Original analysis, Plot 1. Reported zeros are included; missing rates are excluded. The set of reporting countries changes over time.

The sharp fall for America in 2004 required a separate check. The number of reporting countries rose from 2 to 15, with no countries lost. When we restricted both years to the United States and Canada, the median rose from 17.8 to 18.6 pmp. The apparent decline therefore reflects the entry of countries with lower rates, rather than a fall among the original reporters.

Table 1. America reporting-coverage check

Comparison	2003	2004
All reporting countries	2	15
Median across all reporters, pmp	17.8	3.1
Median for the same two countries, pmp	17.8	18.6

This is a two-year comparison using the same countries. It does not create a balanced panel for the full 2000–2024 period. Elsewhere, small regional samples also limit interpretation: Africa’s 2024 median is based on four reporting countries.

The COVID stress test and recovery

To examine the pandemic period, we compared each country's deceased donor rate with its 2019 level. With the corrected population data, the original analysis includes 70 countries with valid rates in 2019, 2020 and 2021 and a positive 2019 baseline. In this sample, 58 countries (82.9%) recorded a decline in 2020, and the median change was -21.4%. The contraction was widespread, although some countries recorded increases.

In my independent extension, I then distinguished a rebound from the 2020 low from a return to the 2019 level. Among the 69 countries retained in the recovery analysis, the median change from 2020 to 2021 was +4.1%. The median gap relative to 2019 was still -18.0%, and 60 countries remained below their 2019 rate in 2021.

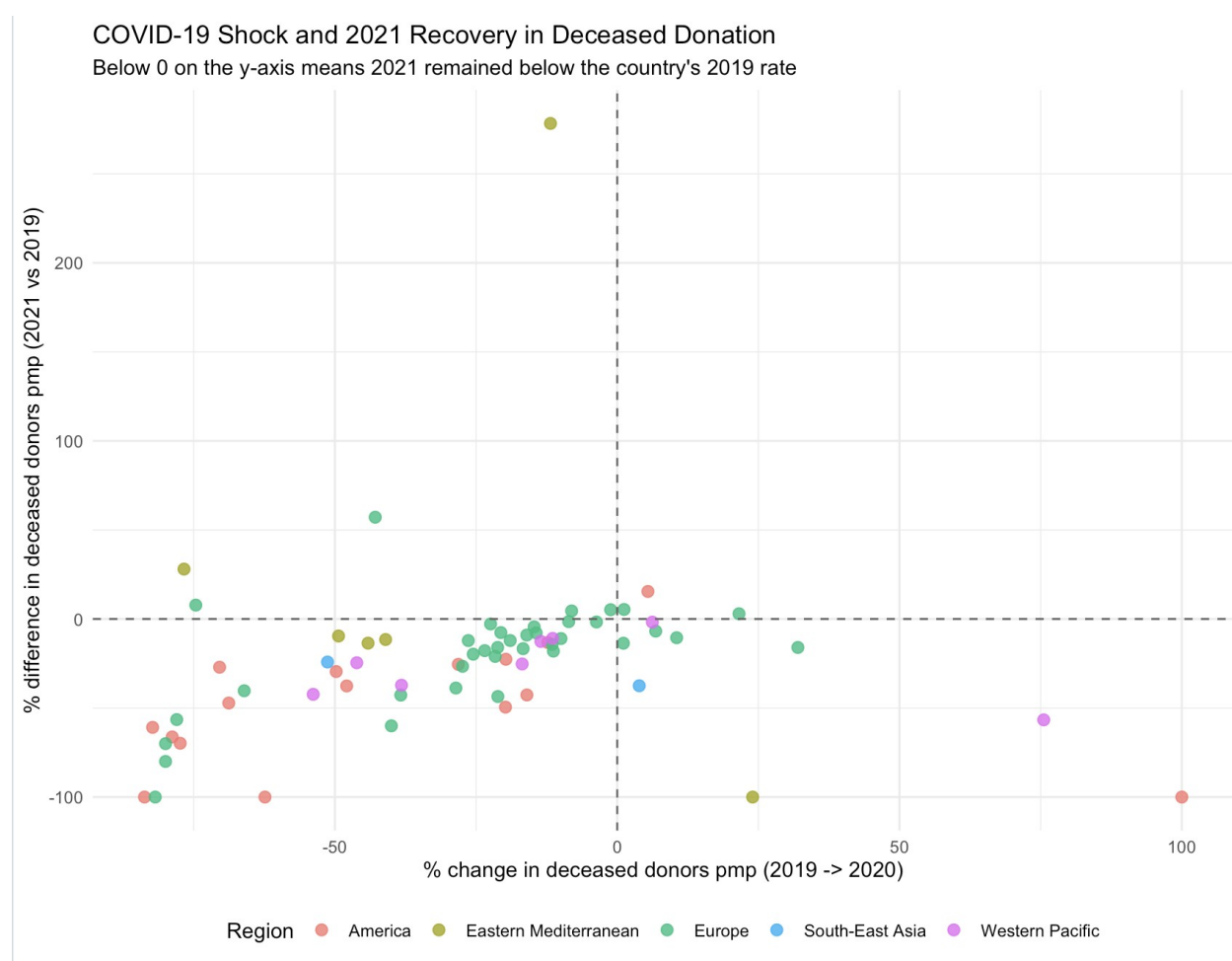


Figure 2. The 2020 change and the remaining gap in 2021, each measured relative to the country's 2019 rate. Independent extension, Plot 3B; 69 countries. Each point is a country. Points below the horizontal zero line remained below their 2019 rate.

The regional summary shows a median 2021 gap of -44.9% for America (16 countries) and -13.6% for Europe (37 countries). These are summaries of the selected countries. They do not measure the pandemic's causal effect or explain whether ICU capacity, transport, staffing or other factors drove the changes.

Sample note. Plot 3B excludes Guatemala because its 2020 rate is zero, making a percentage rebound from 2020 undefined. Its 2021 gap relative to 2019 is still calculable, so this exclusion is a limitation of the current recovery code. The 70-country shock sample and the 69-country recovery sample should not be treated as identical.

Living versus deceased donation

We also examined the source of kidney transplants. For each country and year, the living-donor share is the number of living-donor kidney transplants divided by the sum of living- and deceased-donor kidney transplants. The regional series in Plot 4 takes the arithmetic mean of these country shares, using countries that report both components and a positive total.

The 2024 results show clear differences in donor composition. The mean living-donor share is 76.0% in the Eastern Mediterranean and 68.4% in South-East Asia, compared with 30.8% in Europe. Coverage is uneven, so the sample size is reported alongside each value.

Table 2. Mean country living-donor share of kidney transplants in 2024

Region	Mean share	Reporting countries
Africa	89.1%	4
America	51.1%	22
Eastern Mediterranean	76.0%	9
Europe	30.8%	42
South-East Asia	68.4%	4
Western Pacific	59.8%	9

Each country contributes one share to the mean, regardless of its transplant volume. The table therefore does not give the proportion of all regional kidney transplants supplied by living donors. A high share describes the mix of donor sources; it does not establish weak deceased-donor infrastructure. The share can change because living donation rises, deceased donation falls, or both change at different rates.

Donor types and the regional DCD extension

The original group analysis also compared donation after brain death (DBD) with donation after circulatory death (DCD). My independent extension, Plot 6B, calculates regional DCD shares using only country-years that report both types and have a positive combined total. It divides the regional DCD count by the combined DBD and DCD count. Larger donor programmes therefore carry more weight in this measure.

Among the included reporters, DCD accounts for 31.8% of the combined donor count in America in 2024 and 25.7% in Europe, compared with 15.6% and 16.1% in 2019. Matching the two components within each year improves the comparison, but the reporting countries still change across years. These results describe the composition of reported donation; they do not isolate the effect of clinical protocols or establish that DCD was more resilient to COVID.

Deceased donor utilization and missing data

The original group analysis considered the ratio of utilized to actual deceased donors. Under the GODT definitions, an actual donor has had at least one organ recovered for transplantation; a utilized donor has supplied at least one organ that was transplanted. The ratio therefore measures utilization at the donor level. It does not measure the proportion of all recovered organs transplanted, donor identification or overall system efficiency [2].

Plot 7B shows an apparent fall from 91.5% in 2017 to 77.9% in 2018. However, its numerator and denominator are summed separately, allowing different countries to contribute to each. In 2018, China and Iran reported 6,302 and 923 actual donors respectively but no utilized-donor count. Canada reported 763 utilized donors without an actual-donor count. These gaps make the unpaired aggregate unsuitable for interpreting a change in performance.

Table 3. The 2018 utilization break under different calculations

Calculation	2017	2018
Ratio of separately summed totals — Plot 7B	91.5%	77.9%
Ratio of totals from countries reporting both measures ¹	93.8%	92.9%
Median country ratio, at least 10 actual donors — Plot 7C	96.3%	96.9%

¹ Portfolio validation check: within each year, both totals are restricted to the same countries. The country set can still differ between years. Values are recalculated from the source workbook.

My independent extension adds the median country ratio in Plot 7C and a coverage measure in Plot 8B. The median avoids combining unmatched totals, but answers a different question because countries receive equal weight. In 2018, 86 of 88 actual-donor reporters also reported utilization (97.7%). Yet the two missing countries accounted for 7,225 actual donors. High coverage by country count can therefore conceal a substantial gap in donor volume.

Conclusion and limitations

The analysis shows substantial regional differences in reported donation, an incomplete recovery by 2021, and different mixes of donor sources. It also shows the value of checking coverage before interpreting a trend. The observed sample may be selective, and population adjustment does not account for differences in need, demographics or clinical practice. The conclusions concern the countries and years observed in this dataset; they do not establish the causes of national differences.

Sources and analysis references

[1] GODT source workbook, GODT_downloads.xlsx, Database sheet, 2000–2024. The analysis uses the project’s supplied snapshot. Original project materials: 01-DATASET_GROUP_WORK_2_expanded.R and 02-DATA_ANALYSIS.pdf. Numerical checks use the corrected population import; figure and plot references retain the R script’s labels.

[2] Global Observatory on Donation and Transplantation. [2020 Global Report](#), Materials and Methods, definitions of actual and utilized deceased donors. Used for terminology; the report’s results are calculated from the project workbook.